

E commerce Sales Performance Analysis

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns


# Set random seed for reproducibility

np.random.seed(42)


# 1. Generate synthetic e-commerce sales data

n_rows = 1000

categories = ['Electronics', 'Clothing', 'Home & Kitchen', 'Beauty', 'Sports']

dates = pd.date_range(start='2025-01-01', end='2025-12-31', freq='D')


data = {

    'Order_ID': [f'ORD {1000 + i}' for i in range(n_rows)],

    'Order_Date': np.random.choice(dates, size=n_rows),

    'Product_Category': np.random.choice(categories, size=n_rows, p=[0.3, 0.25, 0.2, 0.15, 0.1]),

    'Quantity': np.random.randint(1, 6, size=n_rows),

    'Unit_Price': np.random.uniform(10, 500, size=n_rows)

}


df = pd.DataFrame(data)

df['Sales'] = df['Quantity'] * df['Unit_Price']

df['Profit'] = df['Sales'] * np.random.uniform(0.1, 0.4, size=n_rows)


# Format floats
```

```
df['Unit_Price'] = df['Unit_Price'].round(2)

df['Sales'] = df['Sales'].round(2)

df['Profit'] = df['Profit'].round(2)


# Save to CSV as required

df.to_csv('ecommerce_sales_data.csv', index=False)


# 2. Inspect data as required by guidelines

print("--- Dataframe Head ---")

print(df.head())

print("\n--- Dataframe Info ---")

df.info()


# 3. Data Preparation for Visualizations

df['Order_Date'] = pd.to_datetime(df['Order_Date'])

df['Month'] = df['Order_Date'].dt.to_period('M')


# Monthly Sales Summary (for line chart)

monthly_sales = df.groupby('Month')['Sales'].sum().reset_index()

monthly_sales['Month'] = monthly_sales['Month'].astype(str)


# Category Sales Summary (for bar chart) - sorted order as required

category_sales = df.groupby('Product_Category')['Sales'].sum().reset_index()

category_sales = category_sales.sort_values(by='Sales', ascending=False)


print("\n--- Monthly Sales Summary ---")

print(monthly_sales)

print("\n--- Category Sales Summary ---")
```

```
print(category_sales)
```

```
# 4. Plotting
```

```
# Configure default styles to prevent truncation and ensure readability
```

```
plt.rcParams["figure.figsize"] = (10, 6)
```

```
plt.rcParams["figure.autolayout"] = True
```

```
# Line Chart: Monthly Sales Trend
```

```
plt.clf() # Clear any existing plots
```

```
plt.plot(monthly_sales['Month'], monthly_sales['Sales'], marker='o', color='b', linewidth=2)
```

```
plt.title('Monthly Sales Performance Trend (2025)', fontsize=14, fontweight='bold')
```

```
plt.xlabel('Month', fontsize=12)
```

```
plt.ylabel('Total Sales ($)', fontsize=12)
```

```
plt.grid(True, linestyle='--', alpha=0.6)
```

```
plt.xticks(rotation=45)
```

```
plt.savefig('monthly_sales_trend.png', dpi=300)
```

```
# Bar Chart: Sales by Product Category (Sorted)
```

```
plt.clf() # Clear plot for the next chart
```

```
bars = plt.bar(category_sales['Product_Category'], category_sales['Sales'], color='teal',  
edgecolor='black')
```

```
plt.title('Sales Performance by Product Category', fontsize=14, fontweight='bold')
```

```
plt.xlabel('Product Category', fontsize=12)
```

```
plt.ylabel('Total Sales ($)', fontsize=12)
```

```
plt.grid(axis='y', linestyle='--', alpha=0.6)
```

```
# Adding value labels on top of the bars for clarity
```

```
for bar in bars:
```

```
    yval = bar.get_height()
```

```
plt.text(bar.get_x() + bar.get_width()/2, yval + 2000, f"${yval:,.2f}", ha='center', va='bottom',  
fontsize=10)
```

```
plt.savefig('category_sales_performance.png', dpi=300)
```

OUTPUT:

```
9:11 m 5G 66%
colab.research.google.com/drive/1vUOk4oHhFu85woC1XfuR_1lbFarVYyr#scrollTo=rOdPVPp1l6n
Untitled8.ipynb
category_sales_performance.png , upi=300/

--- Dataframe Head ---
Order_ID Order_Date Product_Category Quantity Unit_Price Sales Profit
0 ORD1000 2025-04-13 Clothing 4 246.87 987.48 203.89
1 ORD1001 2025-12-15 Electronics 5 433.14 2165.69 250.35
2 ORD1002 2025-09-28 Clothing 4 298.48 1193.91 130.76
3 ORD1003 2025-04-17 Electronics 2 193.91 387.82 88.04
4 ORD1004 2025-03-13 Electronics 5 150.03 750.17 94.05

--- Dataframe Info ---
<class 'pandas.core.frame.DataFrame'>
```

```
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colab.research.google.com/drive/1vUOk4oHhFu85woC1XfuR_1lbFarVYyr#scrollTo=rOdPVPp1l6n
Untitled8.ipynb
category_sales_performance.png , upi=300/

--- Dataframe Info ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 7 columns):
# Column Non-Null Count Dtype
---
0 Order_ID 1000 non-null object
1 Order_Date 1000 non-null datetime64[ns]
2 Product_Category 1000 non-null object
3 Quantity 1000 non-null int64
```

```
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colab.research.google.com/drive/1vUOk4oHhFu85woC1XfuR_1lbFarVYyr#scrollTo=rOdPVPp1l6n
Untitled8.ipynb
category_sales_performance.png , upi=300/

2 Product_Category 1000 non-null object
3 Quantity 1000 non-null int64
4 Unit_Price 1000 non-null float64
5 Sales 1000 non-null float64
6 Profit 1000 non-null float64
dtypes: datetime64[ns](1), float64(3), int64(1), object(2)
memory usage: 54.8+ KB

--- Monthly Sales Summary ---
Month Sales
```

```
9:12 m 4G 66%
colab.research.google.com/drive/1vUOk4oHhFu85woC1XfuR_1lbFarVYyr#scrollTo=rOdPVPp1l6n
Untitled8.ipynb
--- Monthly Sales Summary ---
Month Sales
0 2025-01 64215.94
1 2025-02 74094.05
2 2025-03 51365.40
3 2025-04 75309.82
4 2025-05 72180.65
5 2025-06 58573.13
6 2025-07 51685.86
7 2025-08 89036.68
8 2025-09 60964.96
9 2025-10 61543.15
```

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Onedoc.pynd

RAM Disk

9 2025-10 61543.15

10 2025-11 54670.15

11 2025-12 75605.74

--- Category Sales Summary ---

Product_Category Sales

2 Electronics 234779.12

1 Clothing 190791.73

3 Home & Kitchen 157437.55

0 Beauty 126136.09

4 Sports 80101.04

Sales Performance by Product Category